



Experimental study on suppression of vortex-induced vibration of bridge deck using vertical stabilizer plates

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ABSTRACT

Despite the wind-resistant design performed during the design stage, an existing bridge may still experience severe vortex-induced vibrations (VIVs) because of the installation of new affiliated facilities during its operation period. Therefore, it is necessary to apply control measures to suppress the VIV induced by such changes in configuration. In this study, the VIV performance of a streamlined box girder bridge during the installation of new auxiliary piping system was investigated through a series of sectional model tests and flow visualization tests. Severe vertical VIVs of the bridge used in this study were observed at lower wind speed ranges during wind tunnel tests. Based on the flow pattern around the sectional model, it was inferred that the flow across the deck bottom surface was entirely separated by the new pipelines, which resulted in the periodic generation of shedding vortices in the wake zone and induced deck vibration. Twin downward vertical stabilizer plates (TDVS) were found to be an effective aerodynamic countermeasure for improving the VIV performance of the bridge and could significantly decrease the amplitude of the vertical VIV motion. Additional flow visualization investigations showed that after the TDVS were installed, the vortex shedding in the wake was mitigated, and the location of the vortex shedding was further away from the deck, and hence, considerably suppressed the vertical VIVs. It was found that the effectiveness of the TDVS in improving the aerodynamic performance was significantly dependent on the TDVS parameters, i.e., the plate height and spacing between plate elements. The optimal values of these parameters were correspondingly identified for the bridge deck used in this study after a comprehensive evaluation.

1. Introduction

Streamlined closed-box girders are widely used as the main girders of long-span bridges owing to their excellent mechanical properties and aerodynamic performance. However, the girders may still experience several aeroelastic problems because they become more flexible and possess decreased damping capability with the increasing bridge span. Among the aeroelastic issues, the vortex-induced vibration (VIV) is a common aeroelastic phenomenon that occurs in flexible structures at lower wind velocities, and it is a significant challenge in the wind-resistant design of long-span bridges. VIV has been observed in many actual long-span bridges and has been analyzed in several early studies (Larsen et al., 2000; Battista and Pfeil, 2000; Frandsen, 2001; Fujino and Yoshida, 2002; Macdonald et al., 2002; Ge et al., 2011; Seo et al., 2013;

Hwang et al., 2019). Though the VIV is an amplitude-limiting vibration and is not as dangerous as the flutter, continuous vibrations with relatively large amplitude may cause the discomfort to passengers and inevitably reduce the fatigue life of bridges (Miyata, 2002; Tao et al., 2017). Hence, when a bridge has the problem of fulfilling the VIV performance requirements, it is necessary to implement effective countermeasures to limit the VIV amplitudes under allowable values or even eliminate the VIV to the minimum extent.

Several countermeasures for the suppression of the VIV of long-span bridges have been proposed in previous studies. Currently, these countermeasures can be generally divided into three categories (Zhou et al., 2018; Hu et al., 2019): structural, mechanical, and aerodynamic countermeasures. Among these countermeasures, mechanical countermeasure achieved using mechanical control devices, such as tuned mass

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dampers (TMDs), are among the most effective mechanical countermeasures for controlling the VIVs of long-span bridges (Weber and Mašlanka, 2012; Morga and Marano, 2014; Xing et al., 2014; Kim et al., 2017). Despite this, aerodynamic countermeasures achieved by adding aerodynamic components to bridge girders are still the most widely used approaches owing to their reliability and simplicity, and these components have been successfully applied to many actual bridges with streamlined box girders (Larsen et al., 2000; Sato et al., 2000; Ge et al., 2011; Kwok et al., 2012; Fujino and Siringoringo 2013; Li et al., 2018). The standard aerodynamic components involved in aerodynamic countermeasures include wind fairings, guide vanes, spoilers, and wind barriers. The generation of large-scale vortices can be minimized by adjusting the flow around the bridge girder using these components, which accordingly, suppresses the VIV of bridges.

The application of three-dimensional geometric disturbances in controlling the VIV has also attracted significant attention. This aerodynamic countermeasure has been proven to be efficient in suppressing vortex shedding, reducing oscillating lift forces, and significantly reducing the drag in bluff bodies (Owen et al., 1999; Darekar and Sherwin, 2001; Bearman and Brankovic, 2004; Dobre et al., 2005). The inherent mechanism of this countermeasure is that vortices separated from the leading edge of bluff bodies are three-dimensional, and disrupting the spanwise correlation of the vortex decreases the vortex-induced force along the spanwise direction, which consequently, decreases the VIV amplitude. Based on this mechanism, two countermeasure approaches are applied currently. The first approach is the spanwise sinusoidal perturbation method (SPPM) (Hangan, 2003; El-Gammal et al., 2007), which mainly involves the application of periodic geometrical distortions on wake generators located at the trailing-edge face, leading-edge face, or both. Studies have shown that the VIV could be successfully minimized by applying the SPPM method. The second approach is the installation of vertical stabilizer plates on the upper, lower, or both surfaces of a bridge deck. Based on the findings of previous investigations on wind tunnel tests (Ge et al., 2009; Diana et al., 2013; Yang et al., 2015; Zhou et al., 2018), this approach can decrease the VIV amplitude, and to some extent, also increase the critical flutter wind velocity for long-span bridges. However, it should be noted that previous studies mainly focused on the control effectiveness of a single vertical stabilizer plate installed at the central position of the bridge girder. The control effectiveness and critical parameters of twin vertical stabilizer plates have not been investigated to date, and previous relevant studies are limited in scope.

Through a series of wind tunnel tests, the main objectives of this study are to investigate the VIV performance of a streamlined box girder bridge during the installation of new auxiliary piping system and to identify the corresponding optimal configuration of twin vertical stabilizer plates (i.e., the plate height and spacing between plate elements). With the increase in pipeline networks, the installation of new auxiliary piping system will become widespread for an existing sea-crossing bridge during its operation period. It is known that a slight modification of a bridge deck section creates a significant difference in the VIV response of the bridge. Therefore, the aerodynamic countermeasure for suppressing the VIV with good economy and engineering applicability should be developed for this bridge type. This study could potentially contribute to the aerodynamic performance evaluation and wind-resistance design of bridges when the new auxiliary piping system are installed.

2. Experimental setup

2.1. Engineering background

Fig. 1 depicts the configuration and cross-section of a cable-supported bridge sited in South Korea. The bridge is two-pylon cable-stayed with a steel streamlined closed-box girder, and it has a total length of 450 m, which comprises a 280 m main span and two 85 m side spans. The main girder of the bridge has a width of 11.7 m and a height of 2.75 m. Two new auxiliary pipes, i.e., the water and gas pipes, were installed on this

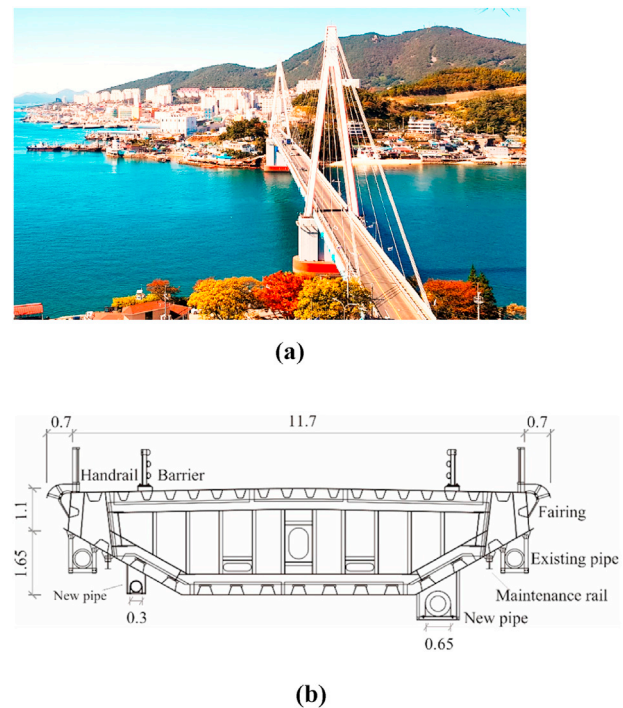


Fig. 1. Description of examined bridge: (a) configuration, and (b) cross-section of girder (unit: m).

bridge to satisfy the growing demand of the urban pipe network. VIV may occur in bridge structures at lower wind velocities owing to the shape changes of the bridge deck. Therefore, it is necessary to investigate the VIV performance of this streamlined closed-box girder by installing new pipelines and developing effective VIV countermeasures.

2.2. Aerodynamic sectional model test

Aerodynamic sectional model tests were performed in an open-circuit wind tunnel at Seoul National University, South Korea. The test section of the wind tunnel had a width of 1 m, height of 1.5 m, and length of 6 m. The wind-tunnel walls of the test section were made of transparent Plexiglas. The wind speed could be continuously adjusted from 0.5 to 23.0 m/s, and the background turbulence intensity was lower than 0.5% of the corresponding wind speed.

The required details of the bridge deck were simulated. The geometrical scale of the section model was set to 1:26 to satisfy the wind-resistance specification requirements and wind tunnel dimensions. Correspondingly, the section model had a width of 0.45 m, height of 0.11 m, length of 0.9 m, and aspect ratio (the ratio of length to width) of 2. The main bodies of the section model and wind fairing were made of balsa wood, and the subsidiary facilities on the deck, including the pipes, maintenance rails, handrails, and barriers, were manufactured using acrylonitrile butadiene styrene plastic. The model was elastically suspended using four pairs of linear springs and a special bracket that could simulate the vertical and torsional modes in two degrees of freedom. Two end plates were installed at both sides of the model to ensure the two-dimensionality of the flow over the model and to prevent the end effects. The additional mass was added to ensure that the mass simulation was generated using steel blocks installed at the bracket. As the VIV is typically sensitive to the damping ratio, viscous dampers (oil dampers) were installed on the system bracket to induce variable damping. The wind speed was monitored using a pitot tube. In addition, the displacement responses of the model were measured using four laser displacement sensors arranged outside the wind tunnel. The sampling frequency of the sensor was set to 500 Hz for a sampling period of 60 s. The section model installed in the test section of the wind tunnel is shown in Fig. 2(a),

and the spring suspension system is depicted in Fig. 2(b). The dynamic parameters of the model system, which were identified using the free vibration method before the VIV tests were performed, are listed in Table 1.

2.3. Flow visualization test

Flow visualization tests using the smoke-wire technique were performed for the section model in the same wind tunnel to investigate the characteristics of flow field around the section model qualitatively. A single smoke-wire manufactured using the nichrome alloy was installed 15 cm upstream of the model leading edge. The wire was installed vertically within the test section through orifices in the walls of the wind tunnel. A wire diameter of 0.3 mm was selected to ensure that the wire did not introduce measurable disturbances into the flow field and to accommodate a sufficient amount of smoke-generating liquid providing the optimum smoke density. Engine oil mixed with paraffin oil was used as the smoke-generating liquid. A variable power transformer was utilized to regulate the voltage supplied to the wire. All the electrical connections were located outside the test section to prevent flow interference. Images of the flow were captured using a high-speed camera, and lighting was provided using a high-intensity light source positioned far downstream of the section model. The camera and lighting device are shown in Fig. 3(a), and the schematic of the experimental setup is illustrated in Fig. 3(b).

3. VIV performance of deck installing new pipelines

3.1. Vibration test results

Vibration tests were performed at five wind attack angles ($\alpha = 0^\circ$, $\pm 1.5^\circ$, and $\pm 3^\circ$) and in two wind directions ($\beta = 0^\circ$ and 180°), as shown in Fig. 4, owing to the asymmetric characteristics of the bridge deck. The attack angle was positive when the nose faced upward.

The vertical and torsional VIV responses of the deck model at the lowest damping ratio (vertical = 0.2%; torsional = 0.15%) were first evaluated at five wind attack angles and in two wind directions in a smooth flow. Fig. 5 shows the test results. The amplitudes and wind speeds were both converted to those of the prototype bridge.

Significant vertical VIVs were observed at five different wind attack angles. A lock-in range appeared at a wind speed range of 10–14 m/s when the wind direction was 0° (Fig. 5(a)). The most unfavorable wind attack angle was -3° , which had a 370 mm peak amplitude; nevertheless, the peak amplitudes of -3° , -1.5° , and 0° were very close (Fig. 5(a)). When the wind direction was changed to 180° , a lock-in range was still observed at a wind speed range of 10–14 m/s (Fig. 5(c)). The peak

Table 1
Parameters for aerodynamic sectional model tests.

Name	Parameter	Unit	Prototype	Similarity ratio	Model
Geometric scale	Length (L)	m	23.40	1:26	0.900
	Width (B)	m	11.67	1:26	0.449
	Height (H)	m	2.75	1:26	0.106
Equivalent mass	Mass (m)	kg/m	7.832×10^3	$1:26^2$	11.558
	Mass moment of inertia (I_m)	kg-m ² /m	1.115×10^5	$1:26^4$	0.245
Frequency	Vertical (first symmetric)	Hz	0.509	–	2.077
	Torsional (first symmetric)	Hz	2.103	–	8.615
Damping ratio	Vertical (ξ_v)	%	–	1	0.20
	Torsional (ξ_t)	%	–	1	0.15

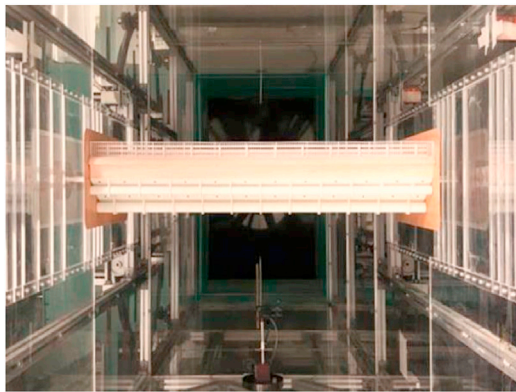
amplitudes of VIVs and the most unfavorable wind attack angle were mostly unchanged relative to the VIVs at a wind direction of 0° . Based on the above comparison, a good agreement between the VIV performances at 0° , -1.5° , and -3° attack angles was observed.

Torsional VIVs also occurred at five different wind attack angles (Fig. 5(b) and (d)), and the corresponding lock-in ranges and peak amplitudes were influenced by the wind attack angles and wind directions. However, the onset wind speeds of torsional VIVs all exceeded 40 m/s, which is rarely reached at the bridge site. The influence of torsional VIVs on the comfort of the bridge was ignored because the traffic was controlled in this case. Hence, only the vertical VIV of the deck are analyzed in detail in this paper, and the torsional VIV was neglected in the following sections. Additional tests were conducted adopting damping ratios of 0.2%–1.0% to determine the vertical VIV performance for different damping ratios. The results are presented in Fig. 6.

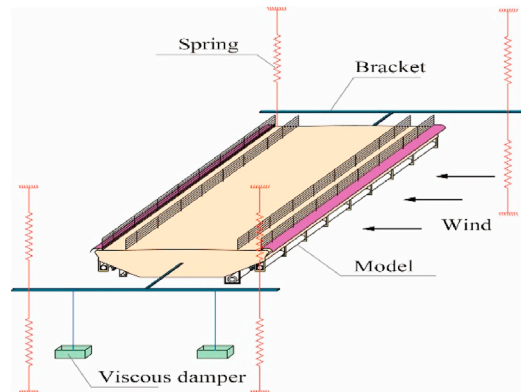
According to the Korean design guidelines for cable-stayed steel bridges (KSCE, 2006), the maximum allowable amplitude of vertical VIV should be lower than 0.5 m/s^2 . By assuming the single-frequency harmonic motion during the vertical VIV, the maximum allowable vertical displacement can be determined as follows,

$$[h_a] = \frac{0.5}{(2\pi f_v)^2} \times 1000 = 48.88 \text{ mm} \quad (1)$$

where f_v is the frequency of the first vertical mode. The vertical VIV responses of the deck at five different attack angles could not satisfy the allowable amplitude unless the modal damping ratio exceeded 1.0%. For steel bridges, the design damping ratio recommended in the Korean code (KSCE, 2006) is 0.4%, whereas 0.36% and 0.5% are specified in the British code (HAE, 2001) and Chinese code (CMT, 2004), respectively.



(a)



(b)

Fig. 2. Sectional model tests: (a) sectional model in wind tunnel and (b) spring suspension system.

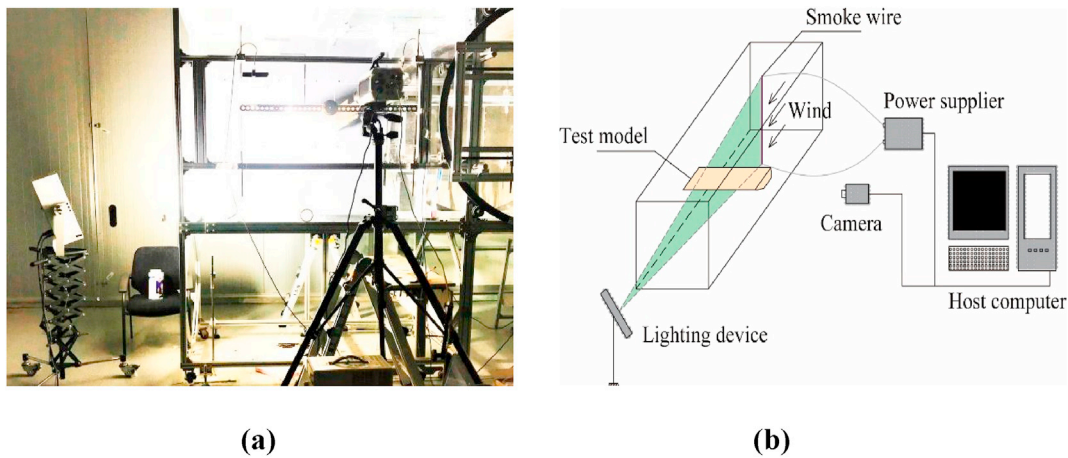


Fig. 3. Flow visualization tests: (a) high-speed camera and lighting device, and (b) schematic of experimental setup.

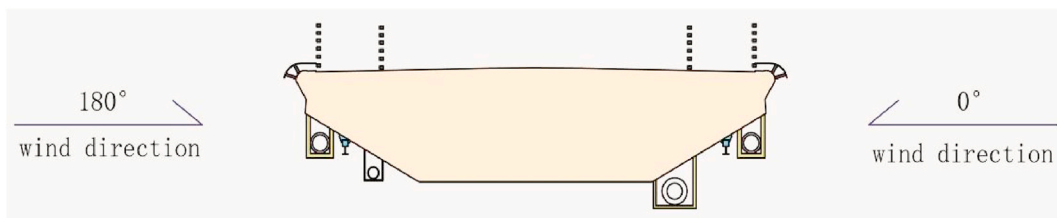


Fig. 4. Schematic of wind direction in sectional model tests.

Therefore, it is difficult to predict that the damping ratio of the actual bridge will exceed 1%. A bridge constructed around the same time and having a similar structural detail with the bridge evaluated in this study showed the measured damping ratio of approximately 0.6% (Kim et al., 2013). Accordingly, the vertical VIV performance of the bridge should be enhanced, and effective countermeasures should be developed to suppress VIVs.

3.2. Characteristics of flow field around deck during installation of new pipelines

Over the past few decades, the flow visualization has been an essential tool for explaining the flow behavior. Although the application of flow visualization cannot yield accurate data of the flow field, it can be used to obtain a qualitative macroscopic description of the overall flow field. When the results of vibration tests and flow visualization tests are combined, highly targeted aerodynamic suppression countermeasures can be developed efficiently.

Fig. 7 shows the flow field of the section model at a wind attack angle of 0° determined using the flow visualization technique. The original deck without new pipes was also subjected to tests for comparison to analyze the VIV triggering mechanism better and determine the component and region of the section inducing the VIV. The results showed that the incoming flow across the deck without new pipes was relatively smooth throughout the entire cycle, and VIVs were unlikely to occur to the original deck. After the new pipes were installed, the flow across the deck was separated from the new pipelines. From $t = 1T/4$ to $t = 4T/4$ of the cycle, a pair of vortices gradually shed from the leeward edge and moved downstream in the wake. The frequency of the vortex shedding at this area approached that of the bridge deck, and thus, the periodic oscillation of the deck (i.e., the VIV), occurred in this case. Because the vortex shedding process was captured through the deck oscillation, the deck movement could have intensified the interaction and energy conversion. This aerodynamic instability is called the lock-in phenomenon. In addition, the large-scale vortices mainly occurred at

the bottom surface of the deck and shed periodically from the lower part of the leeward edge. Hence, it could be inferred that the VIV was mainly caused by the bottom region of the deck. Based on the flow field characteristics determined through flow visualization, the countermeasure using downward vertical stabilizer plates are, therefore, proposed in the next section to suppress the VIV.

4. Suppression countermeasures

4.1. Countermeasure design and tests

In the previous section, the VIV performance of the deck when new pipes are installed, and the flow field around the deck determined through vibration tests and flow visualization tests, are described. Large-scale vortices mainly induced the vertical VIVs of the deck at the bottom region. Based on the positions where the vortices were generated, an aerodynamic countermeasure through the application of downward vertical stabilizer plates was employed. In contrast to previous studies, twin downward vertical stabilizer plates (TDVS) were used in this study.

A series of VIV tests for the deck during the installation of new pipes and the TDVS were performed. Three heights of plate h and four spacings between plate elements d were designed within the range of h/H and d/D ratios (Fig. 8) to investigate the effectiveness of the TDVS on the suppression of the VIV and the effect of critical parameters of the TDVS. The parameter values of the TDVS are listed in Table 2.

Fig. 9 shows the vertical VIV responses of the deck with TDVS installed at four spacings between plate elements for the 0° wind attack angle. The height of the plate was fixed at $0.29 h/H$. Fig. 10 shows the corresponding peak values in these cases, and the case of the deck without TDVS is also included for comparison. From these test results, it was observed that the vertical VIVs were effectively mitigated after the TDVS were installed. The peak values for the deck with TDVS were lower than those without TDVS. When the wind direction was 0° , the four different spacings could significantly improve the VIV performance. Moreover, the peak vertical VIV responses of the TDVS for the different

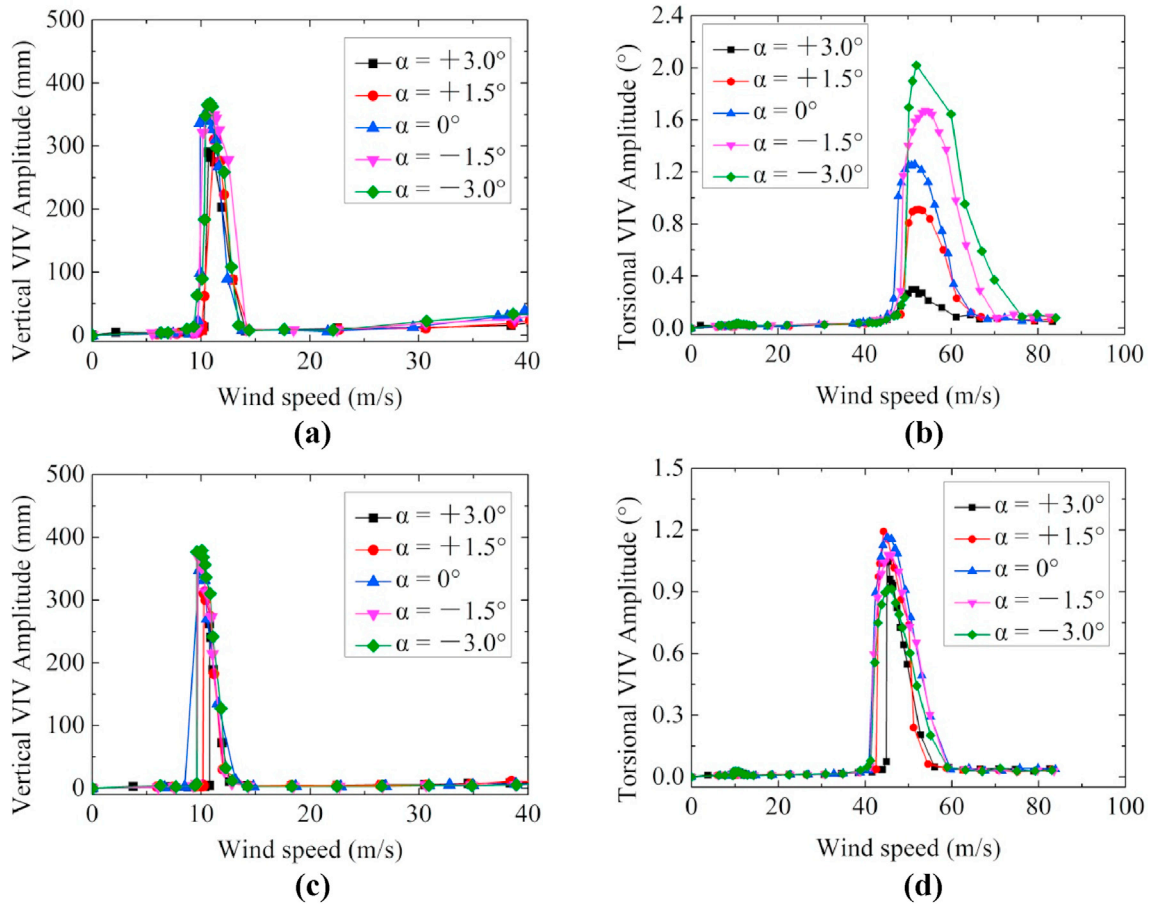


Fig. 5. VIV responses of sectional model installing new pipelines at five angles of attack: (a) vertical response, $\beta = 0^\circ$; (b) torsional response, $\beta = 0^\circ$; (c) vertical response, $\beta = 180^\circ$; (d) torsional response, $\beta = 180^\circ$.

four cases were similar. The TDVS with $0.5 d/D$ exhibited the best control effectiveness. When the wind direction was 180° , the vertical VIV response generally decreased with the increase in the spacing between the plate elements of the TDVS. Therefore, using the TDVS with a $0.5 d/D$ ratio can produce the best VIV control effectiveness in terms of the vertical VIV response among the different spacings between the plate elements of the TDVS.

The vertical VIV responses of the deck when the TDVS were installed at three heights of plates under the 0° wind attack angle were plotted

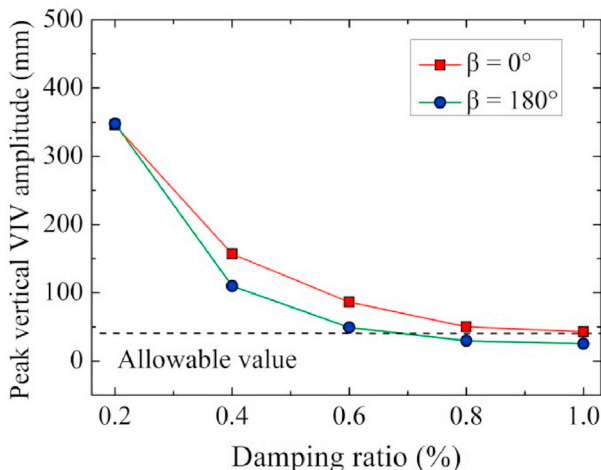


Fig. 6. Variation of vertical VIV response with damping ratio at 0° attack angle.

(Fig. 11). The plate element spacing was fixed at $0.5 d/D$. Fig. 12 shows the corresponding peak values in these cases. It was observed from the results that the vertical VIV responses gradually decreased with an increase in the plate height when the wind direction was 0° . When the wind direction changed to 180° , the VIV responses showed a downward trend with an increase in the TDVS height and then increased after $0.29 h/H$. Therefore, the optimum VIV control effectiveness in terms of vertical VIV response was produced for $0.29 h/H$ of TDVS.

Based on the above results, it can be inferred that the VIVs of the streamlined box girder bridge during new pipeline installation can be efficiently suppressed by mounting TDVS. Additionally, the effectiveness of the TDVS depends on two main parameters of the TDVS, i.e., the plate height and plate element spacing. Hence, optimal parameters for achieving the best control effectiveness of TDVS were obtained in this study. It was found that $0.5 d/D$ and $0.29 h/H$ were the optimal parameters of the TDVS dimensions for suppressing VIV.

Here it should be noted that, though the test results of TDVS with various parameter values generally showed that this aerodynamic countermeasure is effective, the case 1 and case 6 at the 180° wind attack angle did not work, and the wind speed range of VIV occurrence also changed. For 0° wind attack angle, the vortex trails are generated from the big new pipe upstream. By applying the TDVS, the VIV amplitude changes according to the location and size of TDVS, as shown in Figs. 9(a) and Fig. 11(a). However, the Strouhal number is mainly governed by the circular pipe upstream, and the lock-in wind speeds match around 10 m/s for all cases. For 180° wind attack angle, the flow is not disturbed upstream significantly and moves along the streamlined box. Without TDVS, the vortex trails are generated by the big new pipe downstream. In this case, the lock-in wind speed becomes around 10 m/s, as shown in

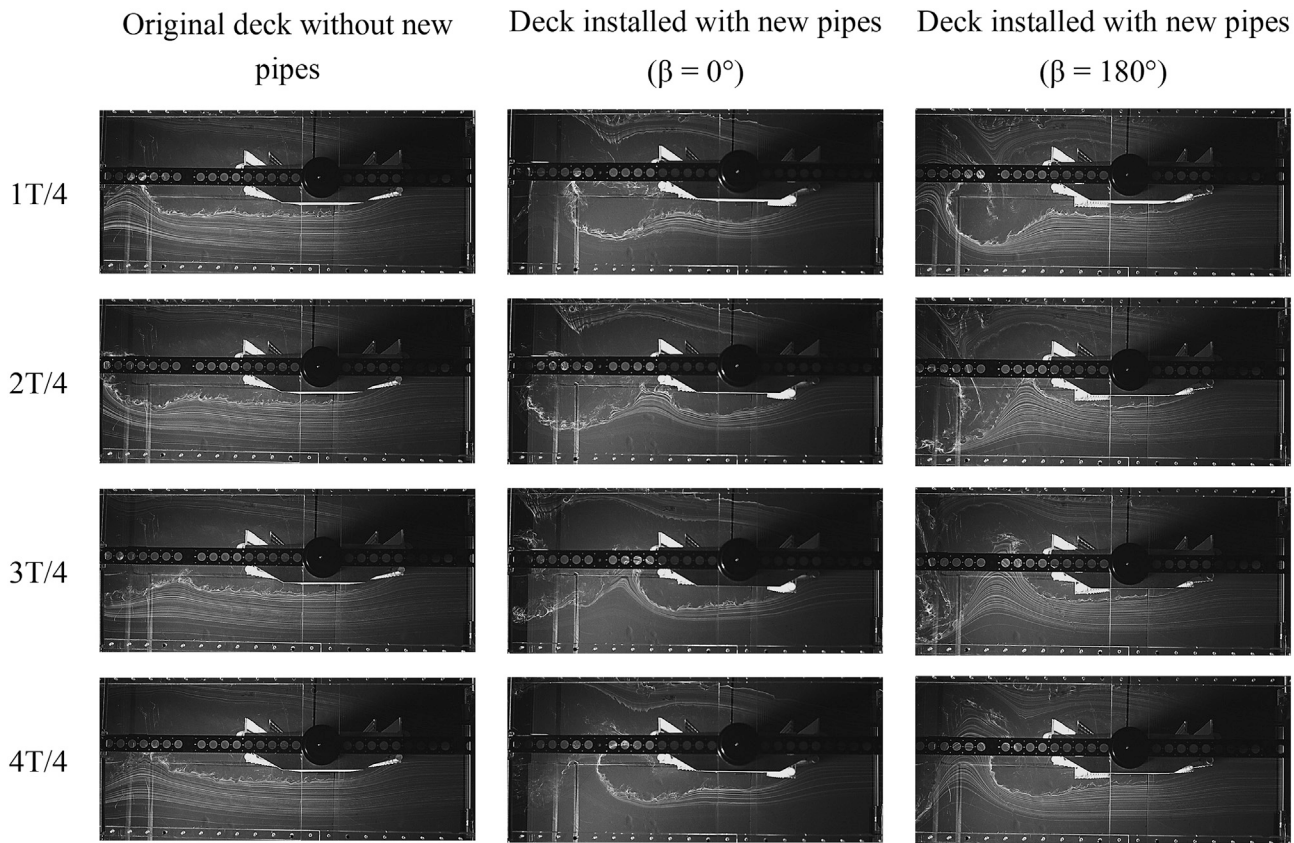


Fig. 7. Flow visualization results for section model at wind attack angle of 0° .

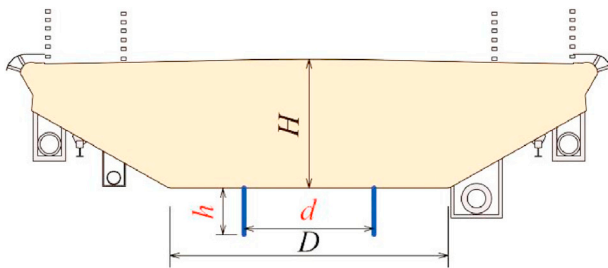


Fig. 8. Schematic diagram of TDVS installed on deck model.

Table 2

Test cases for VIV countermeasures using various TDVS.

Name	Parameter			
	h (mm)	d (mm)	h/H	d/D
Case 1	780	58	0.29	0.25
Case 2	780	116	0.29	0.5
Case 3	780	174	0.29	0.75
Case 4	780	231	0.29	1.0
Case 5	520	116	0.19	0.5
Case 6	1040	116	0.38	0.5

Fig. 9(b). By applying TDVS, we can change the flow stream so that the separated flow zone upstream does not contact the big new pipe downstream. As a result, the VIV of the bridge mitigates, as shown in cases 2–4 in Fig. 9(b). However, when the TDVS is too short, the flow stream comes to contact the big new pipe downstream. Accordingly, we can observe VIV at the wind speed of 10 m/s, like the case 5 in Fig. 11(b). The flow stream changes sensitively according to the position and height of TDVS. In some cases, the flow stream can directly contact the vertical stabilizer

plate upstream, and a different-style vortex trail generates from the vertical plate. The Strouhal number changes and, as a result, we can observe the VIV around 20 m/s, like the case 1 in Fig. 9(b) and the case 6 in Fig. 11(b). Nevertheless, the more detail reason caused for these phenomena may need deeper study due to the complexity of the cross-section used in this study.

In addition, though the onset wind speeds of torsional VIVs all exceeded 40 m/s, and its influence could be ignored because the traffic was controlled in this case, as mentioned before, here it is still necessary to examine the effect of optimal TDVS on the torsional VIVs of the bridge. Fig. 13 shows the torsional VIV responses of sectional model for TDVS with optimal parameters. It can be seen that the optimal TDVS also significantly mitigates the torsional VIVs of sectional model in both wind directions. Hence, the optimal TDVS does not amplify the torsional VIVs of the bridge, but rather works in a better way when the torsional mode is excited. In the next section the characteristics of the flow field around the bridge deck installing the optimal TDVS will be investigated through the flow visualization.

4.2. Characteristics of flow field around deck for installing new pipes and TDVS

Based on the above experimental tests, the VIV performance of the streamlined box girder bridge during the installation of new pipelines significantly improved after the TDVS were fixed, which indicated that such countermeasure would influence the aerodynamic characteristics of the section. When this countermeasure was applied on the bridge deck, the characteristics of the flow field around the section were significantly improved. Hence, the flow field around the cross-section at a 0° wind attack angle in both wind directions were evaluated through the flow visualization in order to investigate the aerodynamic mechanism of the countermeasure. The results are depicted in Fig. 14.

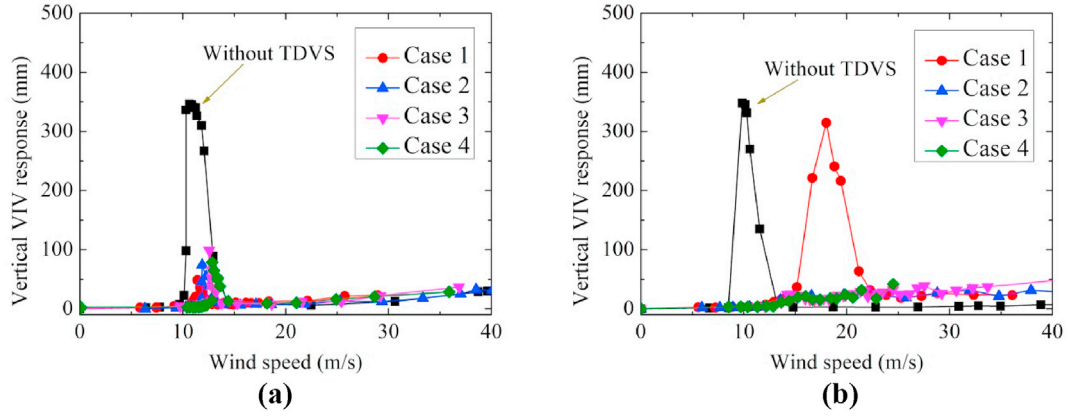


Fig. 9. Vertical VIV responses of sectional model for TDVS installed at four distances of plate element spacing: (a) $\beta = 0^\circ$; (b) $\beta = 180^\circ$.

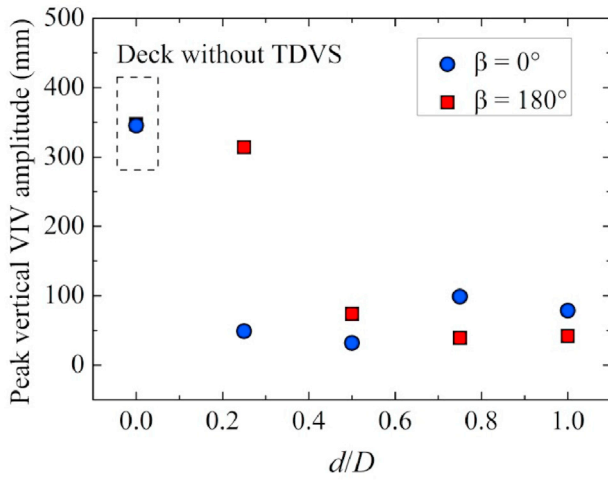


Fig. 10. Peak vertical VIV responses of sectional model for TDVS installed at four distances of plate element spacing.

Based on the analysis described in Section 3.2, the flow on the bottom surface of the deck was entirely separated from the new pipelines, which periodically generated shedding vortices in the wake zone. The results (Fig. 14) indicated that the vortex in the wake zone was relatively stable throughout the entire cycle after the TDVS were installed, and the alternating vortices had disappeared. Moreover, compared to the results for the deck without TDVS, the vortices were further away from the wake, and consequently, the deck was not easily affected by the shedding vortices. Therefore, the VIV of the deck was effectively suppressed when the TDVS was installed.

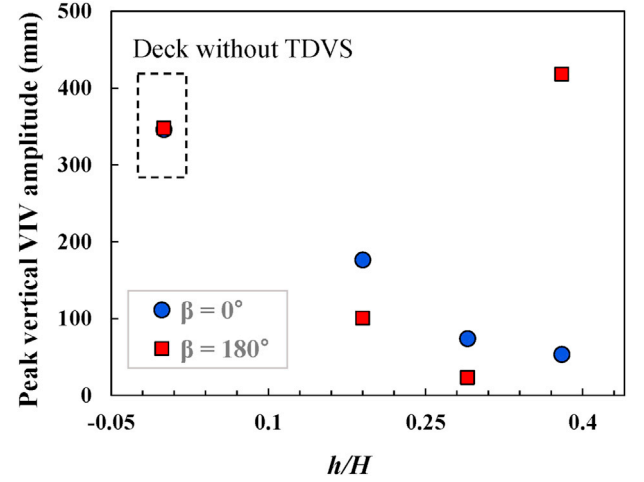


Fig. 12. Peak vertical VIV responses of sectional model for TDVS installed at three heights of plates.

5. Conclusions

Through a series of section model vibration tests and flow visualization tests, the main purposes of this study were to investigate the VIV performance of a streamlined box girder bridge during the installation of new auxiliary piping system and to identify the optimal parameters of the TDVS for suppressing the VIV. The main conclusions of this study are as follows.

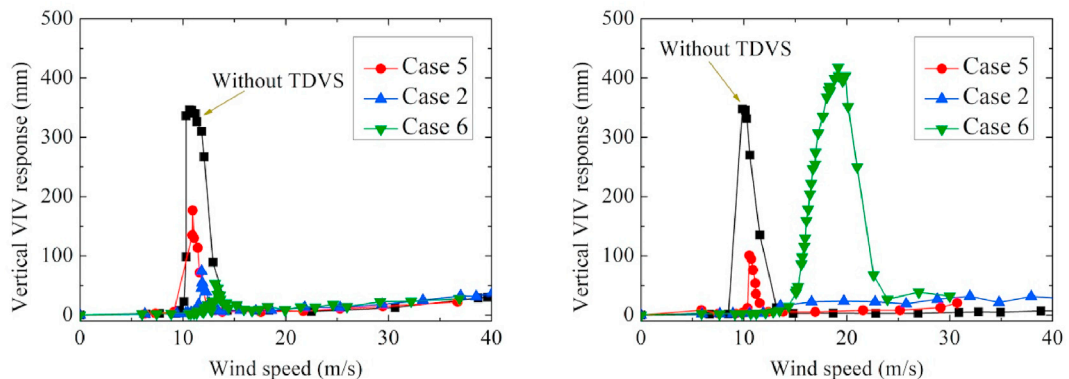


Fig. 11. Vertical VIV responses of sectional model for TDVS installed at three heights of plates: (a) $\beta = 0^\circ$; (b) $\beta = 180^\circ$.

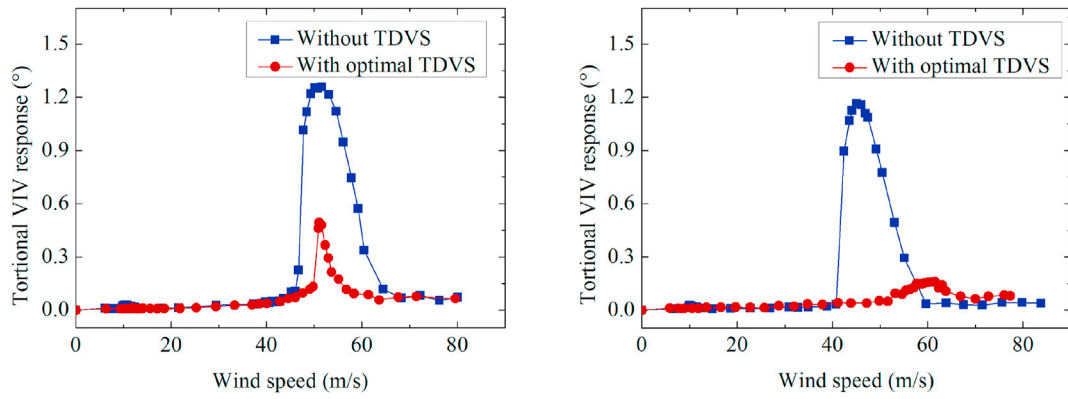
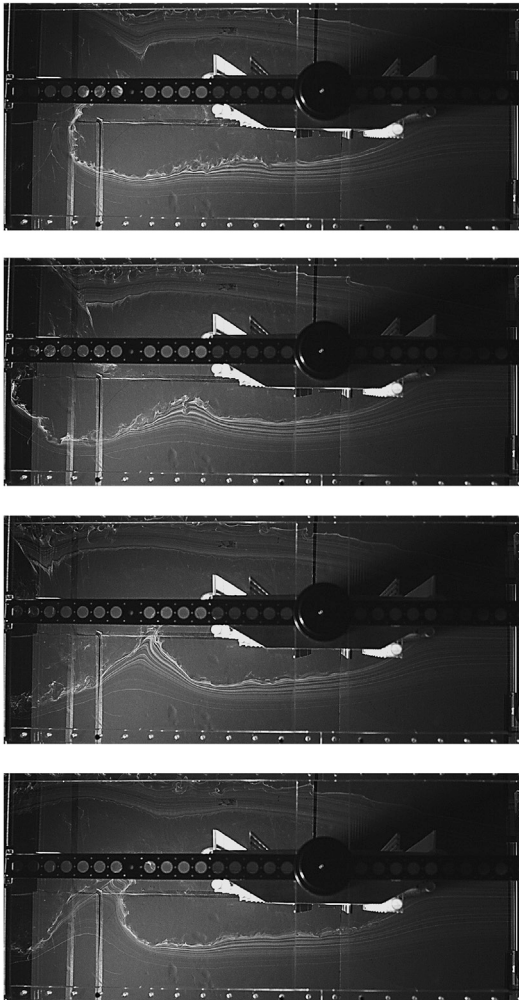


Fig. 13. Torsional VIV responses of sectional model for TDVS with optimal parameters: (a) $\beta = 0^\circ$; (b) $\beta = 180^\circ$.

Deck installing TDVS
($\beta = 0^\circ$)



Deck installing TDVS
($\beta = 180^\circ$)

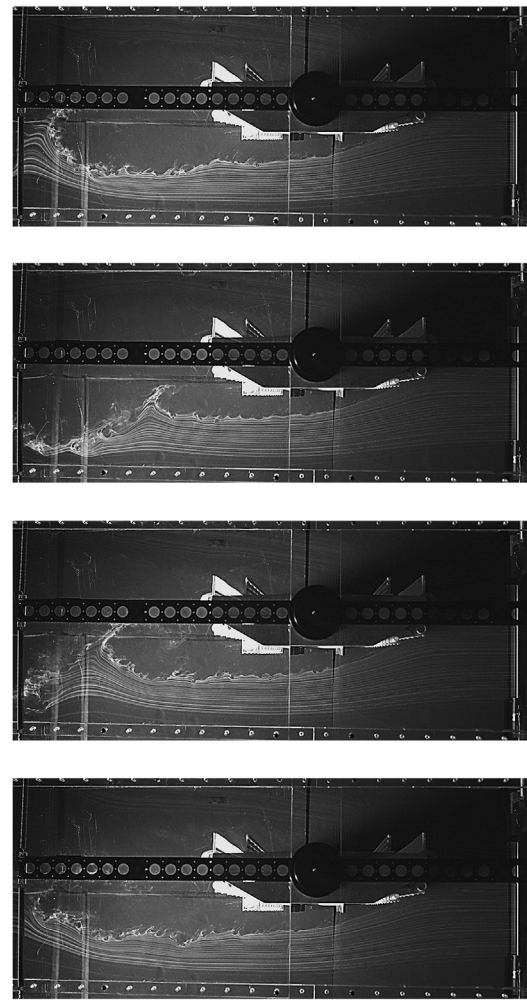


Fig. 14. Flow visualization results for section model for installed TDVS with optimal parameters.

- For a bridge with a streamlined box girder, severe vertical VIVs can occur at lower wind speeds because of the installation of new pipelines, and the VIV properties of the bridge deck vary with the wind direction because of the pipeline asymmetry. The influence of installations of new affiliated facilities on the VIV performance of an

- existing bridge should be considered to ensure the safety and comfort of the users of the bridge structure during the operation period.
- The flow on the bottom surface of the deck was entirely separated by the pipelines, which resulted in the periodical occurrence of shedding vortices in the wake zone and induced the deck vibration. The

installation of the TDVS on the bridge deck bottom could significantly decrease the amplitude of the vertical VIVs of the bridge and was found to be an effective countermeasure for improving the VIV performance of this bridge type.

- The suppression effectiveness of the TDVS depended on two parameters: the plate height and the plate element spacing. For both wind directions, the most effective height and spacing of the plate element for suppressing the VIV were $0.5 d/D$ and $0.29 h/H$, respectively. After the TDVS with these optimal parameters were installed, the large-scale vortices under the bridge deck move further away from the wake zone, and the vertical VIV responses of the bridge significantly decreased at the zero angle of attack, and thus, the requirement for wind-resistant design was satisfied.

CRediT authorship contribution statement

Yang Yang: Conceptualization, Writing - original draft, Methodology. **Soomin Kim:** Investigation, Data curation, Visualization. **Youchan Hwang:** Methodology, Investigation, Validation. **Ho-Kyung Kim:** Conceptualization, Validation, Writing - review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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